

Sentence Classification Is the Wrong Task: Query-Conditioned Verification of Literature-Mined Biotic-Interaction Triples

Anonymous ACL submission

Abstract

Literature-mining systems that populate biotic-interaction databases retrieve candidate (species, relation, species) triples with supporting passages, then filter them with a sentence-level interaction classifier before curation. We show that this filter answers the wrong question. On 100 expert-graded triples retrieved by a deployed system, the filter accepts 91 and reaches precision 0.714, which is not significantly above accepting every candidate (0.650; one-sided exact binomial $p = 0.119$); it rejects 9 of 35 unsupported triples. We decompose those 35 errors into entity-linking (10), predicate (9) and *pair-binding* (16) failures, and show that a perfect per-component oracle still yields precision 0.802, because 16 errors have every component individually correct and are invisible to component-wise validation. Conditioning the decision on the queried triple removes most of the gap without any training: a locally hosted 32B model, prompted zero-shot with the candidate triple, rejects 32 of 35 at precision 0.949 ($p < 10^{-4}$), and reaches F1 0.886 (95% CI [0.846, 0.921]) on a 299-item pair-level benchmark where the best of our supervised models reaches 0.823. We then explain why *supervised* pair conditioning has repeatedly failed in this setting: our paired training corpora contain 606 and 11,602 distinct taxon pairs against 79,460 in the retrieval pool, and the effect of conditioning flips sign with that diversity. We release the graded benchmark, the audited corpora, and a distilled cross-encoder.

1 Introduction

Biotic-interaction databases such as GloBI (?) are populated in part by text mining: a retrieval system proposes candidate triples (s_1, r, s_2) together with the passage that allegedly supports them, and a filter decides which candidates are worth a curator’s attention. The filter is almost always a *sentence-*

level interaction classifier: it is shown the passage and asked whether it describes a biotic interaction.

This paper argues that the sentence-level formulation is a category error for this task, and supports the argument with measurements on a deployed system. Our contributions:

- **The deployed filter does not filter.** On its own expert-graded evaluation set it accepts 91 of 100 candidates at precision 0.714, statistically indistinguishable from accepting all of them (§3).
- **A failure decomposition with a hard ceiling.** Of 35 unsupported triples, 16 have correct entities *and* a correct relation; the triple is wrong only in how the arguments bind. Any component-wise verifier is capped at precision 0.802 (§4).
- **Query conditioning closes most of the gap, zero-shot.** Supplying the candidate triple in the prompt raises rejection from 9/35 to 32/35 and precision to 0.949, and beats every model we trained on a 299-item benchmark (§5).
- **Why supervised pair conditioning failed.** The effect of pair conditioning flips sign with taxon-pair diversity of the training corpus; at 606 distinct pairs the pair slot is noise, at 11,602 it helps by +0.098 AUPRC (§6).

2 Setting

The pipeline. Our retrieval component indexes MEDLINE passages and proposes triples by matching taxon surface forms against an interaction vocabulary derived from the ROBI ontology. Each candidate carries the passage, the two matched taxon surface forms, the matched interaction term, and a heuristic passage score.

System	Acc.	Prec.	Rej.	p
Accept everything	100	0.650	0/35	—
Deployed classifier	91	0.714	9/35	0.119
LLM, pair only	71	0.817	22/35	0.002
LLM, full triple	59	0.949	32/35	$< 10^{-4}$

Table 1: BIOTX100. “Acc.” is candidates accepted, “Rej.” unsupported triples rejected. p is a one-sided exact binomial test against the accept-everything precision of 0.650. The deployed filter is not significantly better than no filter at all.

The graded benchmark. An expert annotator graded 100 retrieved candidates on four axes: whether each taxon surface form was correctly resolved (92/100 and 98/100 correct), whether the relation was correct (90/100), and whether the whole triple was supported (65/100). We refer to this as BIOTX100. A second, larger set of 299 expert-curated items with recovered taxon pairs, drawn from human curation only, is used for pair-level evaluation (TEST299; 54.5% positive).

Evaluation protocol. All decision thresholds are fixed before evaluation and never selected on the reported set; they come from a held-out development split or from a closed-form prior-shift rule. All comparisons report the trivial majority-class baseline, and all intervals are 10,000-sample bootstraps. This matters here: on one of the sources we inherited, an all-positive predictor scores F1 0.919, above every model we tested.

3 The deployed filter does not filter

Table 1 gives the central observation. The deployed classifier accepts 91 of 100 candidates. Its precision, 0.714, is higher than the 0.650 obtained by accepting everything, but not significantly so ($p = 0.119$). It recovers 9 of the 35 unsupported triples. Its recall on supported triples is 1.000 — it rejects nothing that should be kept, and almost nothing that should not.

This is not a threshold-tuning artifact. The classifier’s decisions correlate far more strongly with an axis it was never trained on than with the one it was: **[TARGET: MCC against pair-binding vs. sentence axis]**.

4 What the errors are made of

Table 2 decomposes the failures. The largest class, 16 of 35, consists of triples in which every component is individually correct. Typically the passage

Failure mode	n	Detectable by
Entity linking (s_1 or s_2 wrong)	10	taxon resolver
Predicate (relation wrong)	9	relation classifier
Pair binding	16	joint judgment only
Total unsupported	35	

Table 2: Decomposition of the 35 unsupported triples in BIOTX100. The 16 pair-binding failures have both taxa correctly resolved *and* the correct relation type; the passage simply does not assert that relation *between those two*.

System	F1	95% CI
Trivial (all positive)	0.706	—
Prior multi-task model	0.791	[0.737, 0.840]
Template-trained BiomedBERT	0.821	[0.772, 0.865]
Best supervised (this work)	0.823	—
LLM, query-conditioned, 0-shot	0.886	[0.846, 0.921]
Distilled cross-encoder	[TARGET]	

Table 3: TEST299. Thresholds fixed a priori. The zero-shot query-conditioned model’s interval excludes every supervised system.

asserts the relation between one of the queried taxa and some third organism, or lists several taxa of which only a subset participate.

This bounds any component-wise architecture. A verifier that checks s_1 , r and s_2 separately — however good each check is — rejects at most the 19 component-detectable errors and retains all 16 pair-binding failures, giving precision $65/(65 + 16) = 0.802$. The remaining headroom is reachable only by a decision that considers the triple jointly.

5 Query conditioning

We supply the candidate triple in the prompt and ask a locally hosted 32B instruction model, zero-shot at temperature 0, whether the passage supports it. No training, no in-context examples, no access to the benchmark.

On BIOTX100 this raises rejection of unsupported triples from 9/35 to 32/35 and precision from 0.714 to 0.949 (Table 1). The gain is not merely from naming the two taxa: a prompt giving only the pair, without the relation, reaches 0.817 — so conditioning on the *relation* is worth a further +0.13 precision.

On TEST299 (Table 3) the same model reaches F1 0.886, with a bootstrap interval that excludes every supervised system we trained, including a multi-task architecture developed over several cor-

pus generations.

6 Why supervised pair conditioning failed

Given §5, the natural move is to train a model with the pair in its input. We did, as a segment pair with entity-recognition supervision masked over the query, and the result flips sign depending on the corpus:

Corpus	Distinct pairs	Δ AUPRC
Soft-label, paired	606	-0.003
Repaired v14	11,602	+0.098
Retrieval pool	79,460	—

At 606 distinct pairs the pair slot carries no generalisable signal and costs the model capacity; at 11,602 it helps substantially. The intervention was never wrong; the supervision was two orders of magnitude too narrow. This motivates distilling the query-conditioned teacher over a pair-diverse sample of the retrieval pool ([**TARGET: 20,000 items over 20,000 distinct pairs**]).

7 Related work

[TODO: entity markers and relation classification; shortcut learning and dataset artifacts; biomedical relation extraction; text mining for biodiversity databases. Five verified citations required.]

8 Conclusion

The filter in a deployed biotic-interaction mining pipeline is not significantly better than no filter, because it is asked whether a passage describes an interaction rather than whether it supports the candidate triple. Sixteen of 35 failures are invisible to any component-wise check. Conditioning on the query recovers most of the gap with no training at all.

Limitations

[TODO — unnumbered, after Conclusion, before References, per ARR. Must cover: n=100 and n=299 benchmark sizes; single-annotator gold with no inter-annotator agreement; one retrieval system and one taxonomic domain; LLM contamination cannot be excluded for MEDLINE passages; the distilled student’s dependence on teacher quality.]